

1. Executive Summary

This report presents a comprehensive fraud detection analysis conducted on 50,000 e-commerce transactions spanning 12 months. Using machine learning (Gradient Boosting), behavioral feature engineering, and a custom risk scoring engine, we identified 2,775 fraudulent transactions with an AUC of 0.98 - achieving 83% recall on fraud cases while maintaining 98% precision on legitimate transactions.

50,000	2,775	5.5%	INR 62.3L
Total Transactions	Fraud Transactions	Fraud Rate	Total Fraud Loss

2. Key Findings

- Avg fraud txn (INR 2,246) is 58% higher than avg legit txn (INR 1,419) - high-value txns carry more risk.
- 40.2% of fraud transactions show velocity >= 4 orders/hour - velocity is the #1 behavioral signal.
- Users with prior chargebacks have a 3x higher fraud rate than clean users.
- Peak fraud hour is 9 PM; late night (12AM-5AM) shows 38% higher fraud rate than daytime.
- IP-country mismatch transactions are 2.4x more likely to be fraudulent.
- New accounts (<30 days old) account for 22% of all fraud despite being only 8% of users.
- Credit Card is the most fraud-prone payment method (highest absolute fraud amount).
- Top 3 risky merchants: MERCH232 (9.3%), MERCH224 (9.2%), MERCH179 (8.9%).

3. Model Performance Summary

Model	AUC	Avg Precision	F1 Score	Recall (Fraud)
Gradient Boosting	0.9800	0.8588	0.7531	83%
Random Forest	0.9749	0.7559	0.6450	76%
Logistic Regression	0.9503	0.6193	0.5505	68%

Gradient Boosting was selected as the final model. SMOTE oversampling was applied to handle the 5.5% class imbalance. The top 3 predictive features were: rule_risk_score, prev_chargebacks, and amount_vs_user_avg.

4. Risk Score Segmentation

Risk Tier	Transactions	% of Total	Recommended Action
Low (0-30)	8451	84.5%	Auto-approve
Medium (31-60)	301	3.0%	OTP / 2FA Verification
High (61-80)	115	1.1%	Manual Review

Critical (81-100)	458	4.6%	Auto-Block + Alert
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5. Business Impact & Cost-Benefit Analysis

Based on the test set results (10,000 transactions):

- True Positives (Fraud Caught) : 461 -> INR 1,035,305 saved
- False Positives (Legit Blocked) : 206 -> INR 292,409 revenue lost
- False Negatives (Fraud Missed) : 94 -> INR 211,103 undetected loss
- Net Business Benefit : INR 742,896 per 10K transactions

Annualised (50K transactions/year) : INR 3,714,479 net fraud savings

6. Recommendations

- IMMEDIATE: Auto-block all transactions with risk score > 80 and alert the fraud team.
- PRIORITY: Implement real-time velocity checks - flag users with 4+ orders/hour for 2FA.
- MERCHANT: Place MERCH232, MERCH224, MERCH179 under enhanced monitoring.
- POLICY: New accounts (<30 days) should require phone verification for orders > INR 3,000.
- SYSTEM: Deploy the Gradient Boosting model in production via a REST API scoring service.
- REVIEW: Retrain model quarterly with new labelled data to combat fraud pattern drift.
- REPORTING: Set up a live Power BI dashboard refreshed every 15 minutes for ops team.

7. Appendix - Top 10 Risk Features

Feature	Importance	Description
rule_risk_score	0.142	Composite rule-based risk score
prev_chargebacks	0.118	Number of prior chargebacks
amount_vs_user_avg	0.104	Amount deviation from user's own average
merchant_risk	0.091	Historical fraud rate of merchant
velocity_1h	0.087	Order count in last 1 hour
amount_log	0.079	Log-transformed transaction amount
failed_attempts	0.071	Failed payment attempts before success
user_risk	0.065	Historical fraud rate of user
ip_mismatch_flag	0.058	IP country does not match billing country
account_age_days	0.051	Age of user account in days